American Computer Science League

2023-2024 ● Contest 4: Shorts Problem Solutions ● Senior Division

1. Graph Theory

The adjacency matrix raised to the 2nd power for paths of length 2 is as follows:

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1` & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

The result is squared again for paths of length 4.

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{bmatrix}^2 = \begin{bmatrix} 1 & 1 & 2 & 3 & 4 & 4 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 3 & 1 & 1 \\ 0 & 0 & 0 & 6 & 5 & 5 \\ 0 & 0 & 0 & 5 & 6 & 5 \\ 0 & 0 & 0 & 5 & 5 & 6 \end{bmatrix}$$

There are 15 paths that originate at vertex A (1+1+2+3+4+4=15).

B. 15

2. Graph Theory

By inspection, there are 10 of them. They are: AEA, ABA, ABEA, ADBA, ADBEA, BEDB, BCDB, AEDBA, BDB, BCB

D. 10

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3. Digital Electronics

The digital circuit translates to: $(\overline{AB} + B) \oplus \overline{B + C}$.
It simplifies to:
$= (\overline{AB} + B) * \overline{\overline{B + C}} + \overline{\overline{AB} + B} * \overline{B + C}$
$= (\overline{A} + \overline{B} + B) * (B + C) + \overline{\overline{AB}} * \overline{B} * (\overline{B} * \overline{C})$
$= 1 * (B + C) + AB * \overline{B} * (\overline{B} * \overline{C})$
$= B + C + 0 * (\overline{B} * \overline{C})$
$= B + C$

A. $B + C$

4. Digital Electronics

The digital circuit translates to: $((A \oplus B) + (B \oplus C)) * \overline{C}$. It simplifies to:
$= ((A \oplus B) + (B \oplus C)) * \overline{C}$
$= (A\overline{B} + \overline{A}B + B\overline{C} + \overline{B}C) * \overline{C})$
$= A\overline{B}\overline{C} + \overline{A}B\overline{C} + B\overline{C}\overline{C} + \overline{B}C\overline{C}$
The ordered triples that make this TRUE are: (1,0,0), (0,1,0), (*,1,0). That's 3 since one overlaps so there are 5 that make the circuit FALSE.

D. 5

5. Assembly Language

This program finds the sum of the proper factors of the number NUM which are 1, 2, 4, 7, 14. 28 is a perfect number so the sum is the same as the number.

C. 28